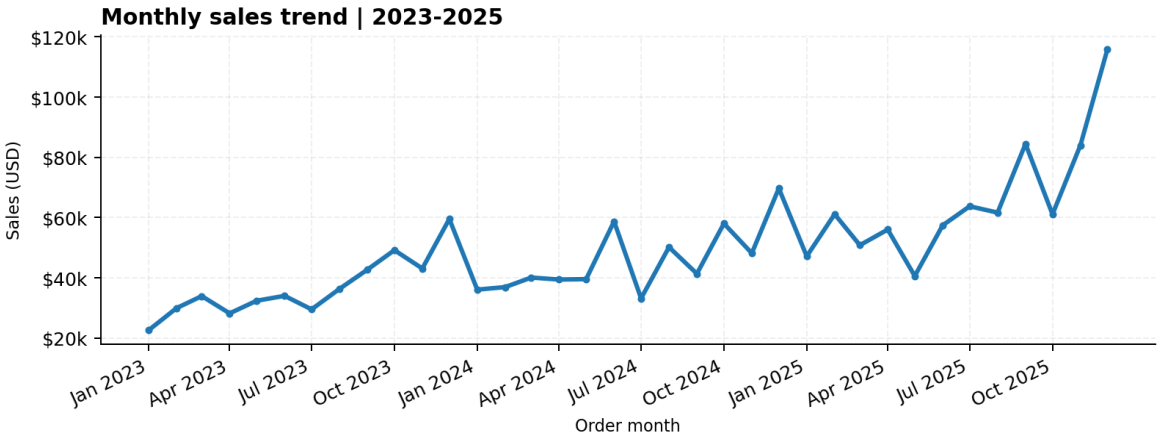
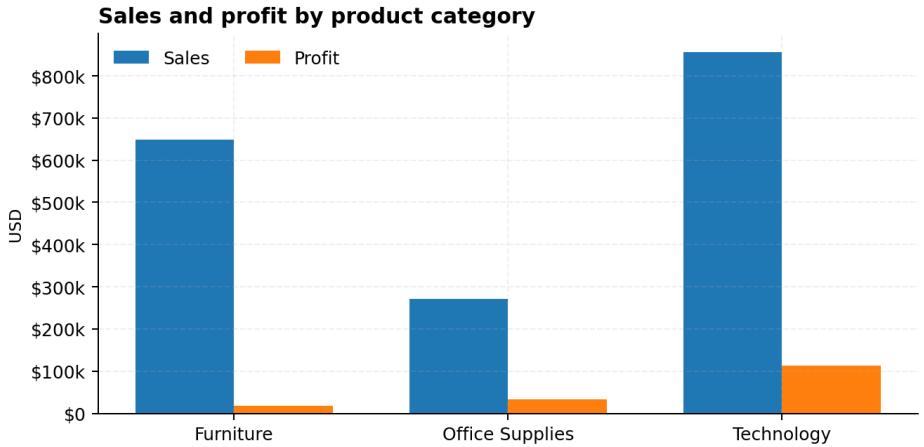
Retail Sales: From Revenue Growth to Profitable Growth

Executive visual report | Sample retail transactions, January 2023 - December 2025 | Currency: USD

$1,776,336	$163,930	9.2%	5,000	699
TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN	ORDERS	CUSTOMERS



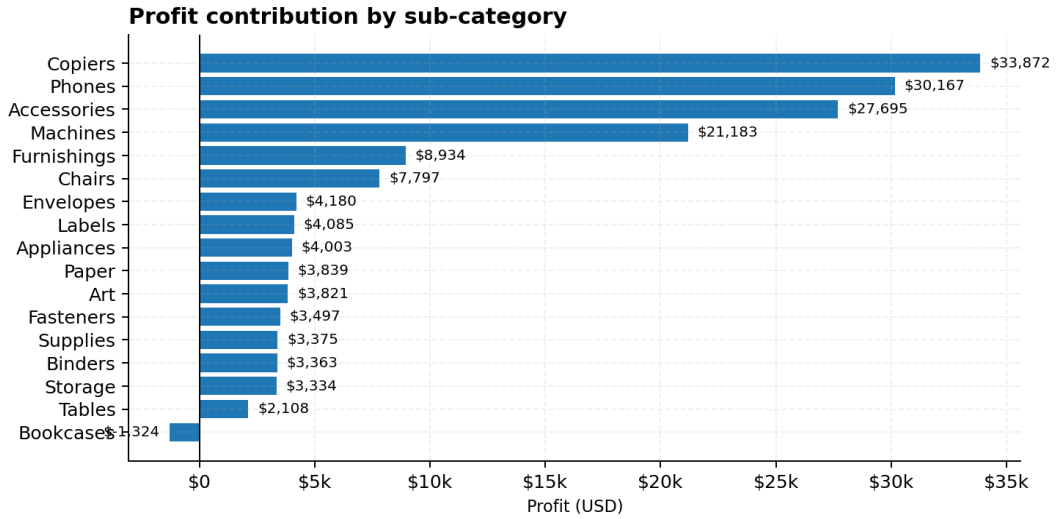


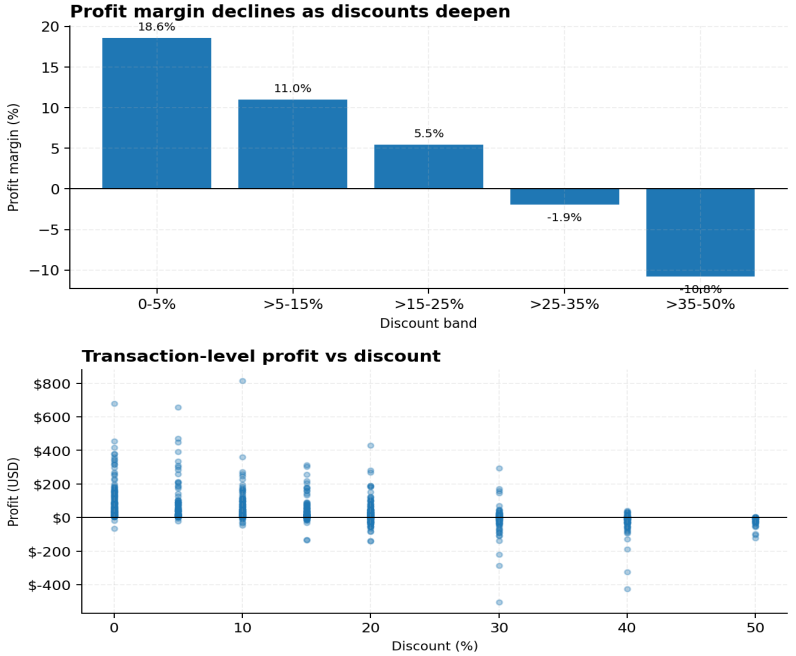
- **Revenue leader:** Technology leads category sales.
- **Profit leader:** Technology generates the highest total profit.
- **Regional leader:** West is the largest sales region.
- **Profit warning:** Bookcases is the weakest sub-category by total profit.
- **Discount signal:** >35-50% is the weakest discount band by profit margin.

Story: The simulated business produces meaningful revenue, but the route to stronger performance is not simply to sell more. Management should protect high-margin categories, investigate loss-making product groups, and make discount decisions using contribution margin rather than revenue volume alone.

Where Profit Is Won - and Lost

The second view shifts attention from topline sales to unit economics and discount quality.





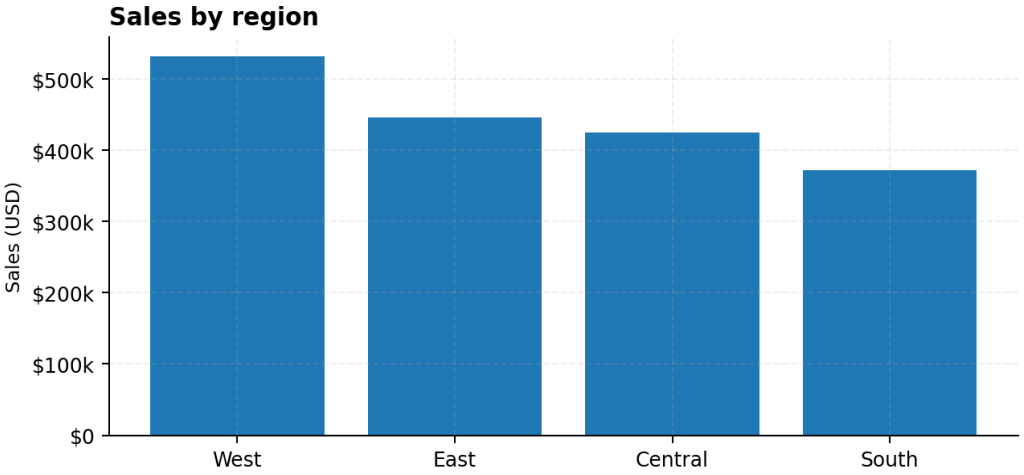
Interpretation: Bookcases records the lowest total profit ($-1,324). The >35-50% discount band has the lowest profit margin (-10.8%). This is an association in the generated data, not proof that discount alone causes the result. Product mix, unit price, and order composition may also matter.

Recommended actions

1. Review loss leaders	Check pricing, supplier cost, freight, returns, and product-level contribution before expanding weak sub-categories.
2. Put guardrails on discounts	Require a margin check or approval for deep promotions; compare incremental volume with profit dollars.
3. Segment performance	Track profit margin by region, customer segment, and product category to avoid blanket decisions.
4. Monitor monthly	Use a recurring view of sales, profit, margin, and discount mix; investigate abrupt changes.

Regional Context, Method, and Caveats

Use context with every chart: metric, unit, timeframe, and the question it answers.



Region	Sales	Profit	Profit margin
West	$532,204	$48,938	9.2%
East	$446,415	$44,701	10.0%
Central	$424,867	$34,172	8.0%
South	$372,850	$36,119	9.7%

Methodology

The raw file contains 5,025 rows. The cleaning workflow removed 25 exact duplicate rows, standardized text whitespace, parsed order and ship dates, converted numeric fields, imputed missing discount values with the median, and labelled missing categorical values as Unknown. The resulting analysis table contains 5,000 rows and 0 missing cells. Sales and profit are aggregated from line items; order count is a distinct count of Order ID. Profit margin is total profit divided by total sales.

Dashboard design choices

Line chart	Monthly sales over time - supports trend and seasonality questions.
Bar charts	Compare categories, regions, and sub-categories with a common baseline.
Scatter plot	Explore the relationship between transaction discount and profit without implying causation.
KPI cards	Provide a compact executive summary before detailed breakdowns.

Important data limitation: This is a synthetic, Superstore-style practice dataset generated for this task. It is not an actual retailer transaction extract and its values/insights must not be represented as real company performance. The generated data covers 2023-2025 and is intended to demonstrate cleaning, chart selection, and business storytelling. For a real business decision, replace it with an approved source dataset and re-run the notebook.

Source note: The assignment brief permits “Superstore.csv or any sales data of your choice.” This project uses the permitted alternative - synthetic retail sales data - because a complete public CSV could not be downloaded in the execution environment.